

Chapter 7: Photosynthesis and Energy Flow

Grade 9 Biology · Unit 3: Plant Systems

7.1 What is Photosynthesis?

Photosynthesis is the biochemical process by which green plants, algae, and certain bacteria convert light energy, typically from the sun, into chemical energy stored in glucose molecules. This process is fundamental to almost all life on Earth, as it forms the base of most food chains and is responsible for producing the oxygen that most organisms depend on for cellular respiration. The overall chemical equation for photosynthesis can be summarized as carbon dioxide plus water, in the presence of light energy, yielding glucose and oxygen.

The process takes place primarily in the chloroplasts of plant cells, specialized organelles that contain a green pigment called chlorophyll. Chlorophyll is essential because it absorbs light most efficiently in the blue and red wavelengths of the visible spectrum, while reflecting green light, which is why most plants appear green to the human eye. Within the chloroplast, photosynthesis unfolds across two interconnected stages, each occurring in a distinct location within the organelle and depending on different inputs and producing different outputs.

7.2 The Two Main Stages

The first stage, known as the light-dependent reactions, occurs in the thylakoid membranes and directly requires sunlight. During this stage, light energy is absorbed and used to split water molecules, releasing oxygen as a byproduct while generating energy-carrying molecules called ATP and NADPH. The second stage, called the light-independent reactions or the Calvin cycle, takes place in the stroma and does not directly require light, although it depends on the ATP and NADPH produced in the first stage. During the Calvin cycle, carbon dioxide from the atmosphere is fixed and progressively converted into glucose through a series of enzyme-catalyzed steps.

- Light-dependent reactions occur in the thylakoid membrane and produce ATP, NADPH, and oxygen.
- Calvin cycle reactions occur in the stroma and use ATP and NADPH to build glucose from carbon dioxide.
- Both stages depend on each other, forming a continuous cycle while light is available.

7.3 Factors Affecting Photosynthesis Rate

Several environmental factors influence how quickly photosynthesis occurs, and understanding these factors is important for both natural ecosystems and agricultural practices. Light intensity,

carbon dioxide concentration, and temperature are the three primary variables that most commonly limit the rate of photosynthesis, often referred to collectively as limiting factors, since at any given moment the factor in shortest supply relative to the plant's needs will restrict the overall rate of the process, regardless of how abundant the other factors may be.

Factor	Effect when increased	Effect when limited
Light intensity	Rate increases until saturation	Rate drops sharply, may halt process
CO ₂ concentration	Rate increases up to a plateau	Calvin cycle slows significantly
Temperature	Rate increases up to optimum (~25-35C)	Enzyme activity slows or stops

Table 7.1 — Summary of major limiting factors in photosynthesis.

7.4 Why It Matters

Photosynthesis is not simply a topic confined to plant biology; it underpins the planet's entire carbon cycle and oxygen supply. Nearly all the energy that flows through ecosystems, from the smallest insect to the largest predator, can ultimately be traced back to photosynthetic organisms capturing solar energy and converting it into a chemical form that other living things can use. Disruptions to photosynthetic activity, whether from deforestation, pollution, or climate change, can therefore have cascading effects across entire food webs and atmospheric composition.